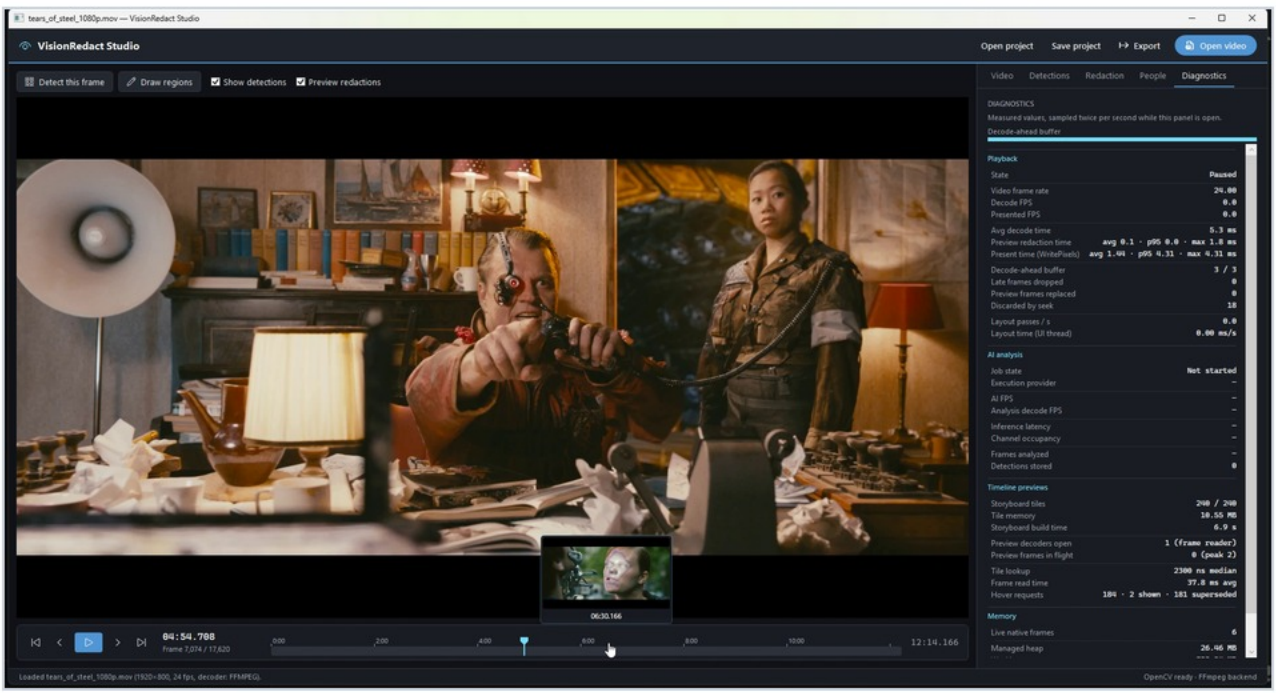
# YouTube-style timeline previews in WPF

A 4K seek can take half a second. Finding a timeline preview takes microseconds. Hover a 12-minute video and see a thumbnail at once; rest, and the exact frame appears.




C# .NET 10 WPF OpenCV FFmpeg

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# Approximate now, exact later

## Hover

the pointer moves



## Nearest tile

shown at once



## Rest 100 ms

the pointer stops

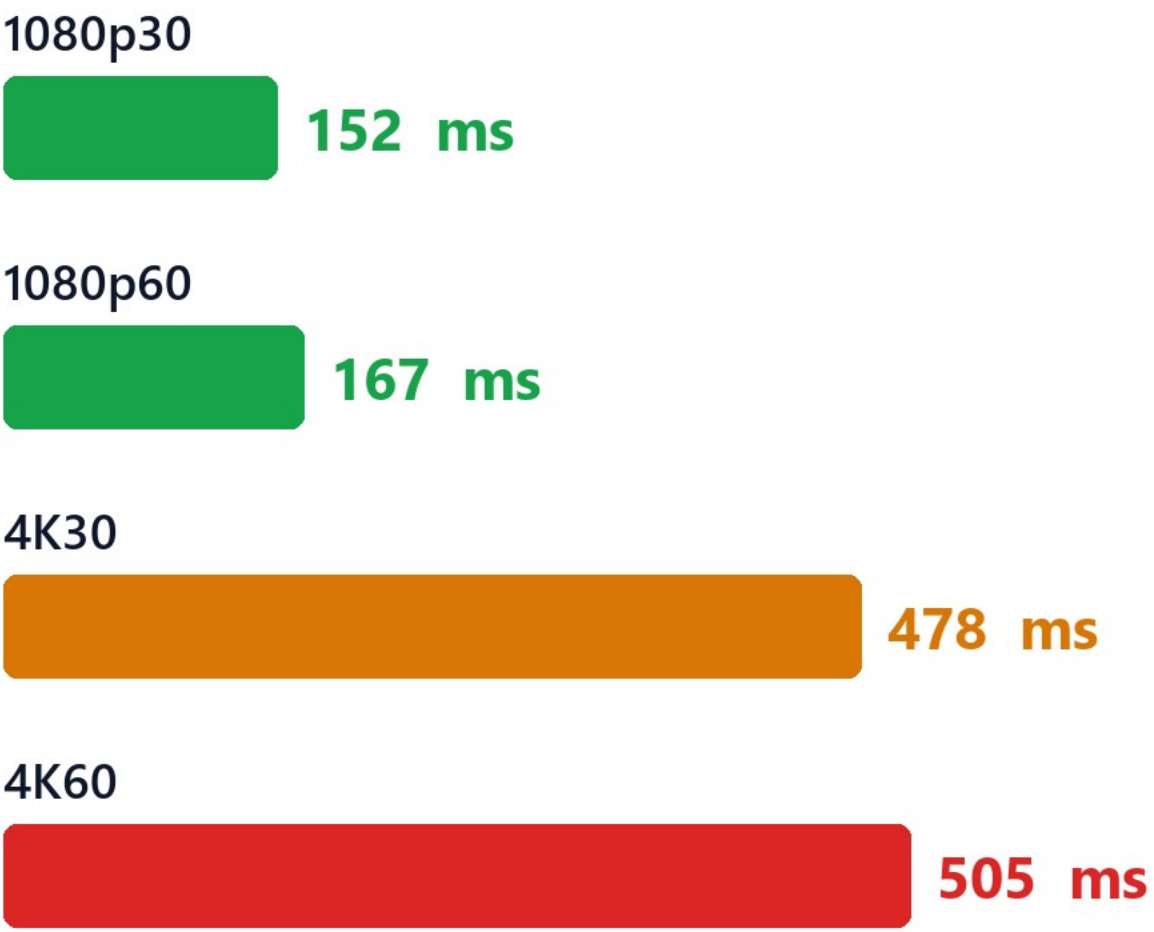


## Exact frame

replaces the tile

# Why not decode on hover?

A seek decodes from the previous keyframe (every 250 frames here).




Median of 40 random seeks. The pointer asks about 32 times a second.

# A storyboard, built in advance

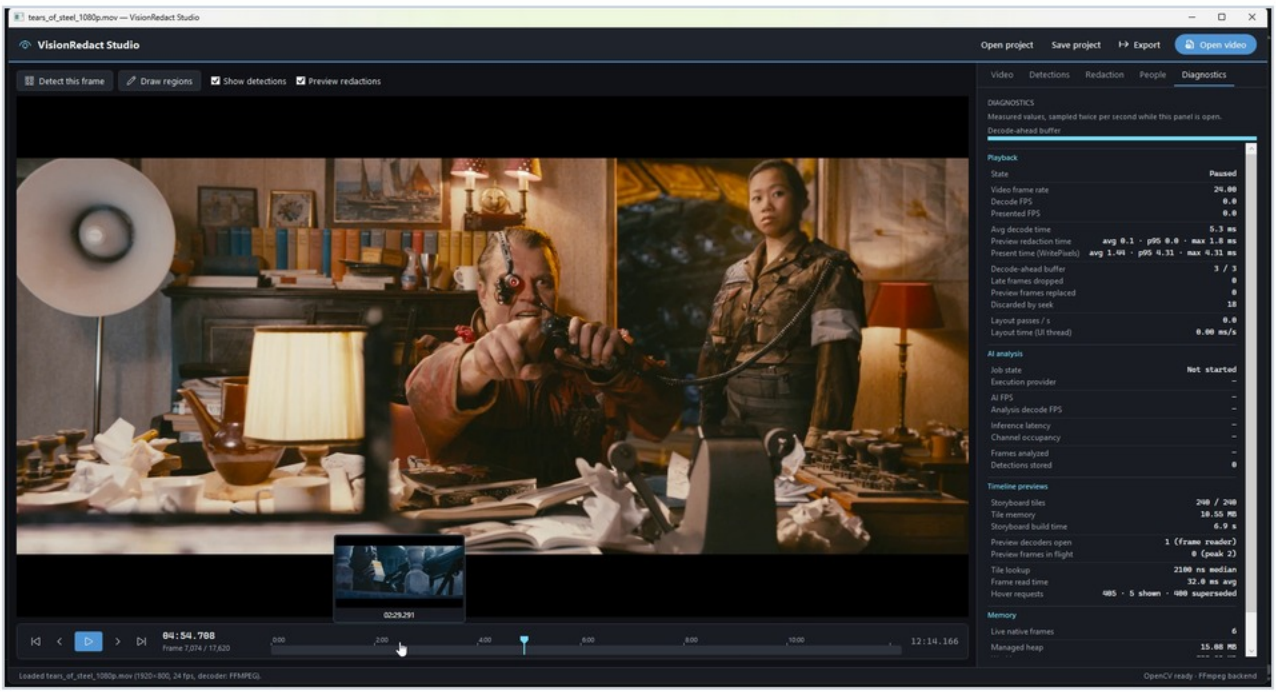
1. **Its own decoder**
   built in the background when a video opens

2. **Tiles at least 2 s apart**
   at most 240, coarse-to-fine (about 3 s apart on the 12-minute film)

3. **Small**
   10.55 MB for a 12-minute film: less than one 4K frame

4. **Fast lookup**
   binary search, about 8 comparisons

# The latest request wins




- Each move cancels the request before it.
- Only a resting pointer gets a decode.
- Over my demo recording: 405 requests, 5 shown, 400 superseded.

# The mean that lied

1. **Benchmark**
   71 ns per lookup

2. **App panel (mean)**
   7.6–11.8 µs

3. **Why**
   one cold first call (~100 µs) dominated

4. **Now**
   median of the last 64: 2–5 µs, at most 0.03 % of a frame

**Report medians for anything with a cold start.**

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# The patterns

1. **Debounce**
   100 ms settle before decoding

2. **Latest wins**
   Interlocked.Exchange + cancellation

3. **Precomputed cache**
   the immutable storyboard

4. **No layout**
   the card moves by a TranslateTransform

# Trade-offs, and what's next

- The storyboard costs a second decoder while it builds.
- Tiles are at least 2 s apart: a 14 s clip gets only 8.
- On 4K files with long keyframe gaps, an exact frame takes about 0.5 s (median).

Next episode

**A YOLO face model running locally in C#, on the GPU.**

Where it's going: the whole pipeline on the GPU.

Footage on slides 1 and 5: Tears of Steel (CC) Blender Foundation | mango.blender.org, CC BY 3.0.